



Birla Institute of Technology & Science, Pilani

Pilani | Dubai | Goa | Hyderabad

ACADEMIC— UNDER GRADUATE STUDIES DIVISION

First Semester 2026-27

Course Handout Part II

Date: 1 August 2026

In addition to part I (General Handout for all courses appended to the Timetable), this portion gives further specific details regarding the course.

Course Number : EEE F111
Course Title : ELECTRICAL SCIENCES
Instructor-in-charge : Hitesh Datt Mathur
[Team of Instructors] : Hitesh Datt Mathur

- 1. Course description:** Course covers basic passive circuit elements, dependent and independent sources, network theorems, circuit analysis techniques, and response of first and second order circuits. Introduction to three - phase circuits, magnetic circuits, transformers, basics of rotating machines. Semiconductors - operation of diodes, zener diodes, bipolar junction transistors and field effect transistors. Biasing techniques and applications of diodes and transistors. Introduction to operational amplifiers and applications. Introduction to Digital Electronics
- 2. Scope and Objective:** The primary objective of this course is to teach the principles of three different aspects of electrical sciences: (1) Circuits Analysis (2) Electronic Devices and (3) Electromagnetics, to the students of mixed disciplines.



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3. Course learning outcomes:

1. Students will learn the circuit analysis (voltage, current and power calculation) techniques for DC and AC (1-phase and 3-phase) circuits using Kirchhoff's current law (KCL), Kirchhoff's voltage law (KVL), Ohms law, Nodal analysis, Mesh analysis, Network theorems (Thevenin's theorem, Norton's theorem, Maximum power transfer theorem, Superposition theorem) etc.
2. Students will be able to perform circuit analysis on electronic circuit consisting of elements such as diode, transistor (BJT, JFET, MOSFET), OPAMP and digital logic.
3. Students will learn construction and operation of electromagnetic devices such as transformer and machines

4. Text books and references:

TB1. Leonard S. Bobrow and Navneet Gupta, Foundations of Electrical Engineering, Oxford University Press, Asian Edition, 2015

RB1. Allan R Hambley, Electrical Engineering: Principles and Applications; 7th Edition (global edition), Pearson education limited, 2019.

5. Course Format & Weekly Workflow

Course work involves 3 lecture hour of class work and 1 tutorial hour engagement in problem solving.

6. Course Plan:

Week No.	Module	Lec.	Coverage	Ref. (TB)	Learning Outcome
1	Basic electrical circuit laws and elements	1-2	Voltage and current sources, basic circuit components, Kirchhoff's current law (KCL), Kirchhoff's voltage law (KVL), Instantaneous power, Inductors, Capacitors	1.1-1.7	Students will learn basic circuit elements and the laws of electrical science to solve basic electrical circuits.
		3	Independent and dependent sources, Source transformation	1.8, 2.4	Students will learn independent and dependent sources, source transformation.



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2-3	Principles of circuit analysis	4,5	Nodal analysis, Mesh analysis	2.1-2.3	Students will be able to solve circuits by nodal and mesh analysis.
		6, 7	Network theorems (Thevenin's theorem, Norton's theorem, Maximum power transfer theorem, Superposition theorem)	2.4,2.6	Students will be able to solve circuits using network theorems.
3-4	Time domain analysis	8-12	Transient response of first and second order circuits (natural and forced response)	3.2-3.5	Students will be able to perform time domain analysis of first and second order circuits.
5-6	AC analysis	13-14	Time-domain analysis, Phasor representation of alternating quantities, j operator and Phasor algebra, Frequency-domain analysis	4.1-4.3	Students will learn concept of phasor, phasor algebra, frequency domain analysis of AC circuits.
		15-17	AC power, Average power, apparent power and complex power, Power factor corrections	4.4-4.5	Students will learn concepts of power in AC circuits.
		18-19	Three phase circuits (Y and Δ connections)	4.6	Students will learn concepts of three phase circuits (Y and Δ connection).
7	Frequency response and resonance	20-21	Filters - Low Pass, High Pass and Band Pass), Resonance and Quality factor	5.1-5.2	Students will apply frequency domain analysis to filter circuits and resonant circuits
8-9	Semiconductor Devices - Diodes	22-24	Basics of PN junction, Junction diode, Ideal diode, Applications in rectifier and clipper	6.2, (partly) 6.3,6.4,	Students will study basics of semiconductors, junction diodes and their use in various electronic circuits.



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		25-26	Zener diode, Applications in voltage regulator and clipper	6.6	Students will learn applications of Zener diodes in electronic circuits.
10	Bipolar Junction Transistors (BJT)	27-29	Brief operation, V-I characteristics of BJT, Circuit analysis	7.1-7.3	Students will learn the operation and analysis of BJT circuits
11	Field-Effect Transistors (FET)	30-32	Brief operation, V-I characteristics of JFET and MOSFET, Circuit analysis	8.1-8.2	Students will learn the operation and analysis of JFET and MOSFET circuits
11-12	Operational amplifier and Digital Electronics	33-34	Basics of operational amplifier and its application, Introduction to Digital Electronics	10.1, 12.1-12.2 (partly)	Students will learn about op-amp circuits and applications.
12-13	Magnetic circuits and transformers	35-36	Analogy between electrical and magnetic circuits, B-H curves	13.1-13.2 (partly), 13.3, 13.4	Students will learn the fundamental concept of electromechanics, magnetic circuits and transformers.
		37-38	Electromagnetic induction, Lenz's law, Transformers (Ideal)	13.6-13.9	Students will learn basic concepts of Ideal transformer.
13-14	Introduction to machines	39-40	Basics of rotating machines	15.1-15.2 (partly)	Students will learn construction and operation of DC/AC machines.



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7. Evaluation Scheme:

Sr. No.	Evaluation Component	Marks (100)	Duration	Date & Time	Nature of Component
1.	Class participation*	5	*	During lecture Hour	-
2.	Quizzes	20	15 Min.	During Tutorial Hour	Closed Book
3.	Mid-Sem. Exam	30	90 Min.	Refer Time Table	Open Book
4.	Comprehensive Exam.	45	180 Min.	Refer Time Table	Closed Book

* Class Participation: This component evaluates student engagement during lecture hours. It will include short quizzes or discussions, conducted throughout the semester during lecture hour. Active and consistent participation will contribute to this component of the overall assessment.

Criterion for NC: A student should obtain 25% of the average marks of the class or 10% of the course total marks, whichever is lower to clear the course. If any student gets marks lower than the prescribed standard mentioned above, he/she may be awarded NC.

8. **Attendance Policy:** Each student is expected to attend all classes and participate in all evaluation components.
9. **Chamber consultation hours:** Displayed on Nalanda (<https://nalanda-aws.bits-pilani.ac.in>).
10. **Makeup Policy:** Makeup will be given only for genuine cases (accompanied by proof) for Mid-sem or comprehensive exam only.

Instructor-in-charge

EEE F111